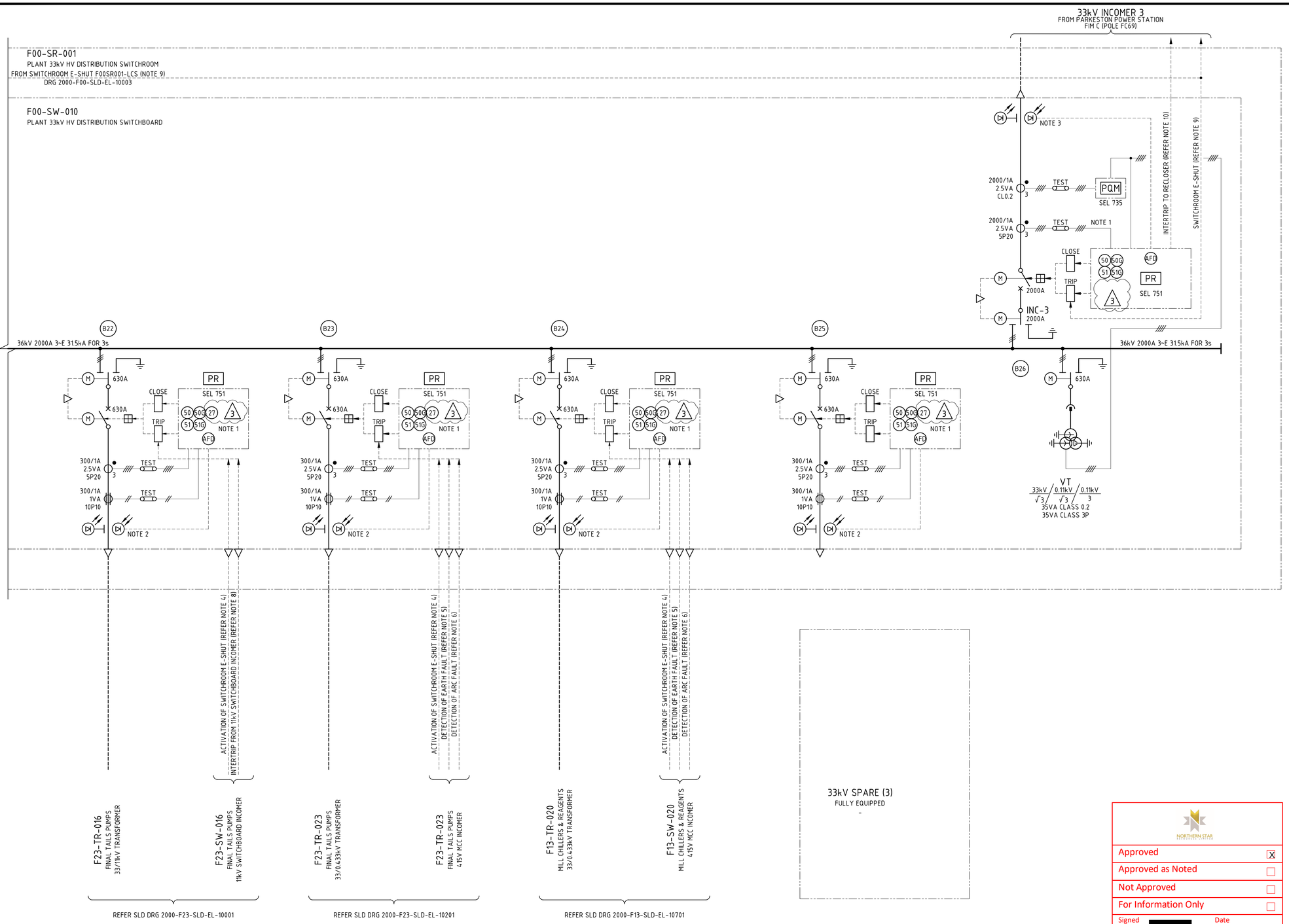


NOTES

1. THE FEEDER PROTECTION RELAY VOLTAGE INPUT SHALL BE WIRED FROM THE CORRESPONDING INCOMER BUS VT.
2. ARC FLASH POINT DETECTORS SHALL BE PLACED IN CABLE ZONES AND CONNECTED DIRECTLY TO THE SEL PROTECTION RELAYS IN EACH PANEL.
3. SWITCHBOARD SHALL BE EXTENSIBLE FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER PER BUS SECTION AS SHOWN.
4. TRANSFORMER 33KV FEEDER SHALL BE TRIPPED ON THE ACTIVATION OF THE SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP THE CIRCUIT BREAKER.
5. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE EARTH FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
6. TRANSFORMER 33kV FEEDER SHALL BE TRIPPED ON THE ARC FAULT DETECTION OF THE DOWNSTREAM MCC INCOMER ACB.
7. TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS SHALL BE HARDWIRED TO THE TRANSFORMER FEEDER PROTECTION RELAY.
8. THE INTERTRIP OF THE UPSTREAM 33kV FEEDER SHALL BE HARDWIRED FROM THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY EITHER ON A TRANSFORMER TRIP, EARTH FAULT DETECTION OR ARC FAULT DETECTION.
9. THE ACTIVATION OF THE PLANT 33KV DISTRIBUTION SWITCHROOM F00-SR-001 EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON VIA A CONTACT HARDWIRED DIRECT TO SHUNT TRIP ALL THREE INCOMER CIRCUIT BREAKERS AND THE CORRESPONDING UPSTREAM POLE MOUNTED RECLOSERS.
10. PROTECTION TRIP FROM THE INCOMER CIRCUIT BREAKER SHALL INTERTRIP THE UPSTREAM POLE MOUNTED RECLOSER.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY NOTES 1 & 2
PQM	POWER QUALITY AND REVENUE METERING
	ARC FLASH SENSOR (POINT TYPE) NOTE 2
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL OR ELECTRICAL INTERLOCK DEVICE
27	UNDER-VOLTAGE
50	INSTANTANEOUS OVERCURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVERCURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
64	EARTH FAULT
67 67G	DIRECTIONAL OVERCURRENT/EARTH FAULT
AFD	ARC FLASH DETECTION

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 24.06.2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

REFERENCES		REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2000-F00-SLD-EL-10055	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD INTERLOCKING SLD SHEET 5 OF 5				SC	AD	JL	MB	DM
2000-F00-DRG-EL-10001	F00-SR-010 33kV PLANT HV DISTRIBUTION SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT	3	30MAY25	FAT REDLINES INCORPORATED	SC	AD	JL	MB	DM
2000-F00-SLD-EL-10005	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 4 OF 5	2	24MAY24	TIER ORDER REVISED.	KR	JL	JL	MB	DM
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM	1	05APR24	TIER ORDER REVISED. REISSUED FOR CONSTRUCTION	KR	JL	JL	MB	DM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX	0	07MAR24	APPROVED FOR CONSTRUCTION	KW	KR	MB	MB	DM

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: GENERAL FIMISTON SITE
F00-SW-010 33kV SWITCHBOARD
SINGLE LINE DIAGRAM SHEET 5

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F00-SLD-EL-10006	3

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELV\F00 - GENERAL FIMISTON SITE\2000-F00-SLD-EL-10006.DWG 03JUN25 6:30AM SANDRA CULL